

Codebasics Digital Asset Assistant

A capstone project brief

Build an AI-powered way for content teams to find, reuse, and trust their assets, and take it end to end, from discovering the real problem to demoing your prototype to stakeholders.

Field	Detail
Project	Capstone 2: Codebasics Digital Asset Assistant
Domain	EdTech / content operations
What you'll build	An AI system that solves the pain point of teams drowning in their own content
Data provided	A dataset (sample files and real public links; more real assets shared during the cohort), see below
You will produce	7 deliverables: user research, opportunity map, PRD, cost estimation, working prototype, presentation, demo video

What This Capstone Is Really Testing

Not just whether you can build an AI app, but whether you can think like an AI PM: start from the real problem, decide where AI genuinely helps, spec it clearly, build a grounded and safe prototype, and tell the story to stakeholders. The build is one step of seven.

1. The Scenario

Codebasics started in 2016 as a YouTube channel by Dhaval Patel and has grown into a learning platform serving learners worldwide through courses, bootcamps, and community programs. A decade of creation leaves a footprint: thousands of assets (course videos, YouTube tutorials, thumbnails, slide decks, PDFs, promotional creatives, scripts) made by multiple teams and scattered across drives, folders, and platforms.

Finding anything is the problem. Naming is inconsistent ("final", "final_v2", "FINAL_USE_THIS"), the same asset lives in three places in four versions, and often only the person who made a file knows where it is, or that it exists at all. A file's name rarely says what's inside it, and nobody can search inside a video or an image anyway.

So teams search, give up, and recreate: a thumbnail that already existed, a deck someone built last year, an intro video re-edited from scratch. The duplication of effort is quiet but constant, and every new course, campaign, and batch adds to the pile. The cost isn't just hours: it's inconsistent branding, outdated versions going public, and creative energy spent remaking instead of making.

2. The Dataset You Receive

You are given a set of sample files you can actually open and links to real, public Codebasics content, more real assets will be shared during the cohort. Note what you are NOT given: any catalog or inventory of assets. None exists; that is part of the problem. Treat "today" as July 2026. A data dictionary (README) is included, read it first.

Source	File	What it is
Sample Assets	sample_assets/ (11 files)	Openable images and PDFs standing in for the thousands of real files; more real Codebasics assets will be shared during the cohort
Public Links	public_links.csv	42 real Codebasics YouTube tutorials, LinkedIn posts, and website pages, published content is part of the library too
Data Dictionary	README.md	Explains every file and column, read it first

3. What You Will Deliver

Take the project through seven stages. Each stage produces one deliverable you submit.

Step 1: Problem Discovery and User Research

- Step into the shoes of a Codebasics team member hunting for an asset. Work through the persona, empathy map, and 5-Whys to get from the surface complaint down to the real root cause.
- Complete the provided activity worksheet.

Deliverable: Filled "2 Template Problem Discovery and User Research.xlsx" (persona, empathy map, 5 Whys, root-cause problem statement, proposed solution).

Step 2: AI Opportunity Mapping

- Break the content lifecycle (create, store, find, reuse) into stages and map where AI could help. Score the opportunities (impact vs feasibility), decide what the solution should and should NOT do, and map each chosen feature to an AI capability and its guardrails.
- Do this on a Miro board: no worksheet is provided for this step, you build the board yourself.

Deliverable: A Miro board (share the link or export a PDF/image).

Step 3: AI PRD

- Write an AI PRD for the solution you chose, using Template PRD.docx.

- Define your North Star metric and the supporting metrics that will tell you whether the solution works. Your PRD must include a dedicated North Star and Metrics section.

Deliverable: An AI PRD document (Word) with a North Star and Metrics section.

Step 4: Cost Estimation

- Before you build anything, estimate what your solution will cost to run: which model(s) you plan to use, expected token usage per interaction, expected volumes, and any infrastructure. Cost is a product decision, not an afterthought, so let it shape your scope.
- Use the provided Cost Estimation.xlsx.

Deliverable: A filled Cost Estimation.xlsx.

Step 5: Build the Prototype (Vibe Code)

- Build a working AI prototype of the solution you specified in your PRD. It must work over the provided dataset (extend it if you need to), produce grounded and trustworthy outputs, and respect the guardrails you defined in Step 2. The architecture and approach are your call, that is the PM's job.
- Use the vibe-coding skills from the course.
- Deploy the solution to cloud using Vercel (free tier) etc.

Deliverable: Working code plus a short README explaining how to run it.

Step 6: Presentation

- Create a slide deck that tells the whole story: the problem and your discovery insights, the opportunity you chose, the solution and a walkthrough, your success metrics, and the risks and guardrails.

Deliverable: A presentation file (.pptx or PDF).

Step 7: Stakeholder Demo Video

- Record a short video (about 5 minutes) demonstrating your prototype as if you are presenting to Codebasics' stakeholders (Dhaval and the team leads). Show the live prototype handling real requests, explain how it stays grounded and safe, and make the business case.
- Speak to the impact, not just the features: you are a PM pitching, not an engineer demoing.

Deliverable: A recorded demo video (share the link).

4. Submission Checklist

#	Deliverable	Format
1	Completed user research worksheet	Excel (.xlsx)
2	AI opportunity map	Miro board link / export
3	AI PRD	Word / PDF
4	Cost estimation	Excel (.xlsx)
5	Working AI prototype	Code + README
6	Presentation	PPTX / PDF
7	Stakeholder demo video	Video link (~5 min)

How You'll Be Assessed

- Problem clarity: did you find the real problem?
- AI fit: is AI the right call, and did you scope it well?
- PRD quality.
- A grounded, safe, working prototype.
- A demo that a real stakeholder would find convincing.

The strongest submissions treat all seven pieces as one coherent story.